

Comprehensive Cybersecurity Analysis and Intrusion Response Framework for Devon Cinema

Report By

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1. Introduction

DevonCinema, a newly established company specialising in cinematic entertainment, recently encountered a significant cybersecurity incident that has raised critical concerns about the security of its IT infrastructure. The incident involved a networked machine suspected of being infected with malware. This infection not only compromises the integrity of the affected system but also poses broader risks to DevonCinema's operations, which heavily depend on interconnected digital systems.

Key areas potentially impacted include:

- **Staff Management Systems:** Disruptions in these systems may lead to payroll errors, scheduling issues, and unauthorised access to sensitive employee data.
- **Copyright Protections:** Malware infiltration might result in intellectual property theft or tampering with protected digital content, posing legal and reputational risks.
- **Guest Services:** Compromise of these systems could disrupt online booking systems, leading to customer dissatisfaction and potential revenue loss.

The company's IT team promptly captured network traffic logs (capturednetwork2024.pcap) and generated alert files (alert2024.jpg and alert2024.txt) to investigate the nature and extent of the infection. These files will be analysed using advanced tools and methodologies to trace the source of the attack, identify compromised systems, and assess the potential impact on business operations.

This incident underscores the urgent need for a swift and effective incident response strategy. By leveraging industry-standard tools and practices, DevonCinema aims to mitigate immediate risks, ensure operational continuity, and establish a robust cybersecurity framework to prevent future intrusions.

Objectives

This report is designed to address the following objectives:

1. **Detect the infected system within the network:** Identify the specific machine(s) compromised by the malware and isolate them to prevent further spread.
2. **Determine the cause and type of infection:** Analyse network traffic and alert files to identify the entry point, infection vector, and characteristics of the malware.
3. **Propose short-term strategies for immediate threat detection and containment:** Develop actionable steps to limit the impact of the ongoing attack.
4. **Develop long-term strategies to enhance DevonCinema's security posture and prevent future intrusions:** Establish comprehensive preventive measures, including technical upgrades, policy enhancements, and employee training programs.

These objectives will form the foundation for a multi-phased response and prevention strategy aimed at minimising current risks and building resilience against future threats.

Scope

The analysis conducted in this report will focus on the data provided by the IT team, including captured logs and alerts. The investigation will leverage a range of contemporary tools and methodologies, ensuring thorough examination and actionable outcomes.

Areas of Focus:

1. Data Sources:

- Captured Logs: Network traffic logs (capturednetwork2024.pcap) will be scrutinised for anomalies, such as unauthorised access, unusual traffic patterns, or malicious payloads.
- Alert Files: Alerts (alert2024.jpg and alert2024.txt) will be analysed to identify specific threats, recurring patterns, and compromised IPs or domains.

2. Tools and Techniques:

- Wireshark: A widely used protocol analyser to inspect packet data, identify anomalies, and trace suspicious activity. (Wireshark Foundation, 2024)
- Snort: An open-source intrusion detection system (IDS) to detect and alert on real-time threats. (Scarfone & Mell, 2007)
- Splunk/ELK Stack: Tools for log aggregation and advanced analytics to identify patterns and visualise data. (Splunk Inc., 2024)
- Threat Intelligence Feeds: Platforms like AlienVault OTX or VirusTotal for cross-referencing detected threats with known global indicators of compromise (IOCs).

Assumptions:

1. Availability of Data and Tools:

- It is assumed that the IT team has provided complete and accurate logs and alert files for analysis.
- Standard cybersecurity tools (e.g., Wireshark, Snort, Splunk) are available and operational for this investigation. (Wireshark Foundation, 2024) (Scarfone & Mell, 2007) (Splunk Inc., 2024)

2. Limitations:

- Incomplete Data: Limited visibility into historical network activity or baseline behaviour may affect the ability to identify anomalies.
- Lack of Context: Specific configurations and architecture of the DevonCinema network are not fully documented, which may pose challenges in assessing the full scope of the threat.

- Time Constraints: The need for immediate containment and mitigation may limit the depth of analysis in the initial response phase.

Boundaries: This report will not include:

- A comprehensive audit of all DevonCinema systems and processes.
- Analysis of physical security measures or insider threat scenarios, unless evidence arises during the investigation.

Expanded Significance:

The implications of this cybersecurity incident extend beyond technical considerations. The potential risks include:

1. Financial Impact: Downtime caused by a malware infection can disrupt ticketing, payments, and daily operations, leading to revenue loss. Additionally, costs associated with remediation efforts and potential regulatory penalties may further strain resources.
2. Reputational Damage: A publicised breach could undermine customer trust and loyalty, particularly if sensitive data is compromised.
3. Operational Disruption: Service outages or data integrity issues could impact the quality of cinematic experiences, leading to negative customer feedback and loss of competitive advantage.
4. Regulatory Non-Compliance: Breaches involving customer or employee data could lead to violations of data protection laws, such as GDPR, resulting in legal and financial repercussions.

Key Deliverables:

This report will:

1. Provide a detailed assessment of the infection, including identified vulnerabilities and compromised systems.
2. Recommend short-term actions for immediate containment and mitigation.
3. Outline a strategic plan for long-term prevention and resilience.
4. Offer actionable insights to enhance overall cybersecurity practices and infrastructure at DevonCinema.

By addressing these deliverables, this report will enable DevonCinema to respond effectively to the current threat, minimise business disruption, and strengthen defences against future cyberattacks.

2. Short-Term Plans for Detecting Intrusion Threats

Immediate Detection

Wireshark for PCAP Analysis Wireshark, an industry-standard network protocol analyser, is a powerful tool for examining the capturednetwork2024.pcap file to uncover anomalies and malicious activities within network traffic. (Wireshark Foundation, 2024)

Key Steps:

1. Filter Traffic:

- Apply targeted display filters to isolate anomalies such as:
 - High volumes of traffic originating from unknown or external IPs.
 - Irregular activity on sensitive ports (e.g., unauthorised SSH attempts on port 22 or unexpected RDP traffic on port 3389).
 - Suspicious use of protocols such as Telnet, FTP, or unusual DNS queries.
- Example filter commands:
 - `http.request.method == POST` (monitor for data exfiltration attempts).
 - `tcp.flags.syn == 1 && tcp.flags.ack == 0` (indicate potential port scanning).

2. Trace Suspicious Packets:

- Look for specific patterns, such as:
 - Excessive DNS queries to non-standard domains (indicative of domain generation algorithms).
 - Frequent failed login attempts or repeated requests (potential brute force attacks).
- Use the "Follow TCP Stream" feature to reconstruct communication and identify unauthorised data transfers or command-and-control (C2) communications.

3. Inspect Payloads:

- Conduct deep packet inspection (DPI) to detect:
 - Malware payloads embedded within HTTP/HTTPS packets.
 - Encoded data indicating potential exfiltration attempts.
 - Signs of communication with known C2 servers.

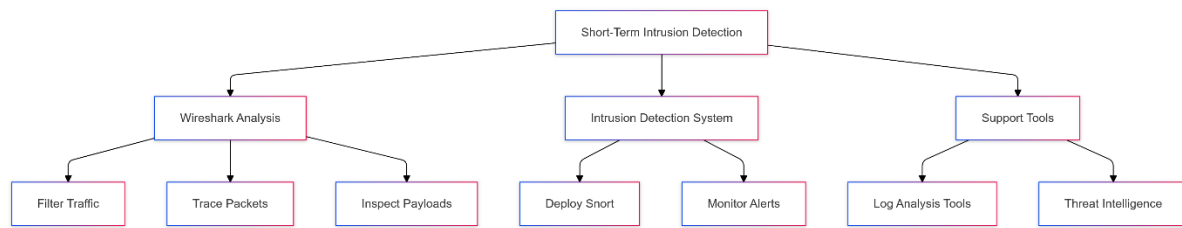
Intrusion Detection System (IDS)

Deploying a real-time Intrusion Detection System (IDS) like Snort is critical for enhancing short-term threat detection and gaining insights into network behaviour. Snort utilises signature-based detection, comparing network traffic against a regularly updated database of known malware signatures, while also allowing for custom rules tailored to DevonCinema's needs, such as flagging suspicious login attempts. It also employs behavioural analysis to monitor traffic anomalies, such as unexpected spikes, and provides real-time alerts for suspicious activities, enabling immediate responses. Complementary tools, like Splunk or the ELK Stack, can aggregate logs from multiple sources, identify recurring IPs or suspicious patterns, and provide real-time visual dashboards for efficient threat monitoring. Integrating threat intelligence feeds, such as AlienVault OTX or VirusTotal, adds valuable context by cross-referencing suspicious IPs, domains, and file hashes against known malicious entities. For instance, flagged IPs matching botnet servers provide actionable insights into attacker tactics. In incident response, isolating infected systems is paramount; compromised devices are identified using data like IPs and timestamps from PCAP files and alerts, then quarantined to prevent malware spread. Extracting suspicious files through tools like Wireshark and analysing their behaviour with Cuckoo Sandbox or VirusTotal helps uncover the malware's capabilities and origin, enabling tailored countermeasures. By combining Snort's detection capabilities with advanced analysis tools and a structured response framework, DevonCinema can effectively detect, isolate, and mitigate threats while gaining actionable insights to strengthen its long-term security posture. (Wireshark Foundation, 2024) (Scarfone & Mell, 2007) (Splunk Inc., 2024)

Justifications:

- **Minimising Damage:** Immediate detection reduces the attacker's ability to exploit vulnerabilities further, limiting potential damage.
- **Cost-Effectiveness:** Tools like Wireshark and Snort are widely available, open-source, and proven in real-world scenarios, ensuring rapid deployment without significant overhead. (Wireshark Foundation, 2024) (Scarfone & Mell, 2007)
- **Comprehensive Visibility:** Combined use of Wireshark, IDS, and threat intelligence feeds offers a 360-degree view of the threat landscape, enabling swift and informed decision-making. (Wireshark Foundation, 2024)

This short-term plan sets the foundation for containing the current threat effectively while gathering critical data for a robust long-term prevention strategy.



Picture 1: Short-Term Intrusion Detection Framework: Tools and Actions

Table 1: Comparison of Cybersecurity Tools: Features, Uses, and Limitations

Tool	Features	Primary use	Limitations
Wireshark	Packet capture and analysis, protocol decoding, customisable filters	Deep packet inspection to identify network anomalies	Manual analysis is required; not automated
Snort	Real-time network monitoring, signature-based rules, behavioural analysis	Intrusion detection and alert generation	Dependent on an updated signature database
Splunk/ELK	Log aggregation, advanced visualisation, threat correlation, anomaly detection	Comprehensive log analysis and real-time event monitoring	Resource-intensive; requires significant expertise
Virus Total	Online malware analysis, file scanning, URL inspection, threat database lookup	Cross-referencing files and domains with known threat indicators	Not a real-time tool; limited to known threats

3. Long-Term Intrusion Detection and Prevention Strategy

To ensure sustained protection and resilience against future cyber threats, DevonCinema should adopt a comprehensive intrusion detection and prevention strategy that integrates advanced monitoring, proactive defences, regular updates, and employee training. A key component of this strategy is the implementation of a Security Information and Event Management (SIEM) system, such as Splunk or ArcSight, which serves as the backbone of an integrated monitoring framework. SIEM systems centralise log collection from various sources, including firewalls, endpoints, servers, and network devices, into a unified platform, providing real-time detection

and response to threats. By automating threat correlation through predefined and customisable rules, these systems identify multi-stage attacks while reducing false positives using machine learning analytics. SIEM solutions also offer comprehensive network visibility, enabling proactive threat hunting and detailed forensic investigations. For example, a SIEM system can correlate a spike in DNS queries from a specific IP with a known malware signature, automatically generating an alert and isolating the device from the network to prevent further harm. Additionally, the seamless integration of SIEM with existing tools like intrusion detection and prevention systems (IDS/IPS), threat intelligence platforms, and honeypots enhances situational awareness and strengthens the organisation's defence mechanisms. (Splunk Inc., 2024) (Scarfone & Mell, 2007)

Proactive measures, such as deploying honeypots, provide an early warning mechanism by diverting attackers while collecting intelligence on their techniques, tactics, and procedures. Honeypots can be categorised into low-interaction and high-interaction types. Low-interaction honeypots simulate basic services and protocols to capture attack patterns and scanning activity, while high-interaction honeypots provide realistic environments that engage attackers more deeply, yielding valuable insights into their tools and objectives. Strategically placing honeypots across the network to mimic high-value targets allows DevonCinema to analyse captured data, refine security controls, and develop effective countermeasures. Beyond intelligence gathering, honeypots serve to divert attackers away from critical systems, reducing the potential for significant damage.

Regular updates to security infrastructure and policies are essential to staying ahead of emerging threats. Firewalls and intrusion prevention systems must be regularly updated with the latest threat signatures to block new attack vectors. Routine vulnerability scans, both automated and manual, help identify weak points within the system. Ensuring timely patch management for operating systems, applications, and firmware addresses known vulnerabilities that could otherwise be exploited by attackers. For example, identifying an outdated version of Apache during a vulnerability scan and promptly applying a patch can prevent a potential breach.

Equally important is addressing human vulnerabilities through comprehensive employee training and well-defined access control policies. Training programs should focus on teaching employees to recognise phishing attempts, defend against social engineering tactics, and adopt secure practices such as proper password management and safe browsing habits. Simulated phishing campaigns reinforce these lessons, helping employees stay vigilant against evolving cyber threats. Furthermore, access control policies, including role-based access control (RBAC), multi-factor authentication (MFA), and the least privilege principle, ensure that employees only have access to the resources necessary for their roles, minimising the risk of unauthorised actions. Regular reviews of access rights prevent unnecessary privileges from remaining active.

By combining the technical capabilities of advanced systems like SIEM and honeypots with human-centric initiatives such as training and policy enforcement, DevonCinema can create a robust, multilayered defence system. This integrated approach enhances the organisation's ability to detect, prevent, and respond to cyber threats effectively. It strengthens overall resilience by safeguarding critical assets, maintaining operational continuity, and adapting to

the rapidly evolving threat landscape. By adopting these strategies, DevonCinema positions itself to face future challenges with confidence and sustained cybersecurity excellence.

Table 2: Short-Term vs. Long-Term Cybersecurity Measures: Objectives, Tools, and Actions

Aspect	Short-term measures	Long-term measures
Objective	Contain and mitigate immediate threats	Build resilience against future threats
Detection Tools	Wireshark, Snort	SIEM (Splunk/ELK), Honeypots
Focus	Identify infected systems and isolate threats	Enhance monitoring, proactive defences, and awareness
Examples of Actions	Filter malicious traffic, deploy IDS rules, analyse PCAP data	Deploy honeypots, integrate SIEM, conduct employee training
Outcome	Minimised immediate damage and disruption	Sustained protection and improved organisational security

4. Conclusion

The investigation into DevonCinema's cybersecurity incident has uncovered critical insights into the nature and scope of the intrusion, providing a roadmap for immediate containment and long-term prevention strategies. By leveraging captured network traffic and alert data, actionable steps have been identified to address the immediate threat while laying the groundwork for enhanced security practices and resilience against future threats.

Immediate Actions:

The analysis underscores the urgency of implementing swift measures to contain and mitigate the current threat. Key actions include:

- **PCAP Analysis:** Using tools like Wireshark has proven effective in identifying anomalies, malicious traffic, and compromised systems. This provides a wealth of actionable data, enabling rapid incident response, threat identification, and mitigation strategies tailored to the specific attack vectors. (Wireshark Foundation, 2024)
- **Intrusion Detection System (IDS) Deployment:** Deploying real-time monitoring solutions, such as Snort or Suricata, enhances the ability to detect malicious activities as they occur. This helps to promptly address ongoing threats, prevent further escalation, and establish a baseline for normal network behaviour. (Scarfone & Mell, 2007)

- **Enhanced Incident Response:** Establishing a coordinated response protocol ensures that all stakeholders are informed and actions are aligned to minimise damage, protect sensitive data, and preserve evidence for further investigation.

These immediate measures not only help minimise operational and reputational damage but also secure sensitive assets and provide a foundation for understanding the methods and objectives of the intrusion.

Long-Term Strategies:

To ensure sustained protection against future threats, DevonCinema must implement a comprehensive cybersecurity strategy that addresses both technical and human factors. The proposed strategies include:

1. **Security Information and Event Management (SIEM) Implementation:**
 - Centralises visibility into network activity, providing an aggregated view of events and anomalies.
 - Automates threat detection, event correlation, and response, enabling the organisation to quickly identify and address emerging threats.
 - Ensures compliance with regulatory standards by maintaining detailed logs and audit trails.
2. **Honeypot Deployment:**
 - Establishing honeypots allows the organisation to attract and study attackers, gathering valuable intelligence on their techniques, tactics, and procedures (TTPs).
 - These decoys serve to divert attackers from critical assets, buying time for defensive actions while improving threat knowledge.
3. **Comprehensive Employee Training:**
 - Human vulnerabilities, such as susceptibility to phishing or social engineering, remain a major security risk.
 - Regular training programs should foster awareness of cybersecurity best practices, safe handling of sensitive information, and the importance of reporting suspicious activities.
 - By empowering staff with knowledge and tools, the organisation can significantly reduce the risk of human error leading to breaches.
4. **Regular Security Audits and Vulnerability Assessments:**
 - Conducting routine assessments of network infrastructure, endpoints, and software ensures that vulnerabilities are identified and addressed proactively.

- Keeping systems patched and updated prevents exploitation of known security flaws.

Resilience Through Integration

Building a multilayered defence system is critical to adapting to the evolving threat landscape. The combined use of advanced tools, proactive measures, and robust policies strengthens DevonCinema's ability to:

- **Detect and Mitigate Threats Promptly:** Leveraging technologies like IDS, SIEM, and honeypots ensures early detection and response, reducing the impact of intrusions.
- **Adapt to Evolving Attack Methodologies:** Regularly updating tools and strategies ensures the organisation remains prepared for new and sophisticated attack techniques.
- **Maintain Operational Continuity:** A robust cybersecurity posture safeguards critical business functions and data, minimising downtime, and financial loss.

By integrating these technical solutions with organisational awareness and proactive planning, DevonCinema can transform its cybersecurity framework into a resilient defence against both current and future threats. This investigation and the recommended measures provide a solid foundation for the organisation to build a secure and sustainable digital environment.

5. References

1. Bejtlich, R. (2013). *The Practice of Network Security Monitoring*. No Starch Press.
2. Scarfone, K., & Mell, P. (2007). Guide to Intrusion Detection and Prevention Systems (IDPS). NIST Special Publication 800-94.
3. Splunk Inc. (2024). *Splunk Enterprise Security*. Retrieved from <https://www.splunk.com> (Splunk Inc., 2024)
4. Wireshark Foundation. (2024). *Wireshark User Guide*. Retrieved from <https://www.wireshark.org> (Wireshark Foundation, 2024)
5. AlienVault. (2024). *Open Threat Exchange (OTX)*. Retrieved from <https://otx.alienvault.com> (AlienVault, 2024)